MSG No. 190.231 $P\bar{6}2'c'$ [Type III, hexagonal]

Table 1: Wyckoff site: `2a`, site symmetry: `32'.`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,3,7,8,9]`
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[4,5,6,10,11,12]`

Table 2: Wyckoff site: `2b`, site symmetry: `-6..`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,2,3,4,5,6]`
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[7,8,9,10,11,12]`

Table 3: Wyckoff site: `2c`, site symmetry: `-6..`

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	`[1,2,3,4,5,6]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{3}{4}\right]$	`[7,8,9,10,11,12]`

Table 4: Wyckoff site: `2d`, site symmetry: `-6..`

No.	position	mapping
1	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[1,2,3,4,5,6]`
2	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[7,8,9,10,11,12]`

Table 5: Wyckoff site: `4e`, site symmetry: `3..`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3]`
2	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[4,5,6]`
3	$[0,\ 0,\ -z]$	`[7,8,9]`
4	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[10,11,12]`

Table 6: Wyckoff site: `4f`, site symmetry: `3..`

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[1,2,3]`
2	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}-z\right]$	`[4,5,6]`
3	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[7,8,9]`
4	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$	`[10,11,12]`

Table 7: Wyckoff site: 6g, site symmetry: .2'.

No.	position	mapping
1	$[x,\ 0,\ 0]$	[1,7]
2	$[0,\ x,\ 0]$	[2,8]
3	$[-x,\ -x,\ 0]$	[3,9]
4	$\left[-x,\ -x,\ \frac{1}{2}\right]$	[4,10]
5	$\left[x,\ 0,\ \frac{1}{2}\right]$	[5,11]
6	$\left[0,\ x,\ \frac{1}{2}\right]$	[6,12]

Table 8: Wyckoff site: 6h, site symmetry: m..

No.	position	mapping
1	$\left[x,\ y,\ \frac{1}{4}\right]$	[1,5]
2	$\left[-y,\ x-y,\ \frac{1}{4}\right]$	[2,6]
3	$\left[-x+y,\ -x,\ \frac{1}{4}\right]$	[3,4]
4	$\left[x-y,\ -y,\ \frac{3}{4}\right]$	[7,11]
5	$\left[y,\ x,\ \frac{3}{4}\right]$	[8,12]
6	$\left[-x,\ -x+y,\ \frac{3}{4}\right]$	[9,10]

Table 9: Wyckoff site: 12i, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[-y,\ x-y,\ z]$	[2]
3	$[-x+y,\ -x,\ z]$	[3]
4	$\left[-x+y,\ -x,\ \frac{1}{2}-z\right]$	[4]
5	$\left[x,\ y,\ \frac{1}{2}-z\right]$	[5]
6	$\left[-y,\ x-y,\ \frac{1}{2}-z\right]$	[6]
7	$[x-y,\ -y,\ -z]$	[7]
8	$[y,\ x,\ -z]$	[8]
9	$[-x,\ -x+y,\ -z]$	[9]
10	$\left[-x,\ -x+y,\ z+\frac{1}{2}\right]$	[10]
11	$\left[x-y,\ -y,\ z+\frac{1}{2}\right]$	[11]
12	$\left[y,\ x,\ z+\frac{1}{2}\right]$	[12]